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An alternative and fast method for determination of glyphosate and aminomethylphosphonic acid (AMPA) residues in soybean using liquid chromatography coupled with tandem mass spectrometry

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A simple and specific method using reversed-phase liquid chromatography coupled with electrospray ionization tandem mass spectrometry (LC/ESI-MS/MS) was investigated, which allowed the determination of residues of glyphosate and its metabolite, aminomethylphosphonic acid (AMPA), in soybean samples. An aqueous extraction with liquid-liquid partition followed by protein precipitation was performed before the LC/MS/MS determination. The quantitation of glyphosate and AMPA was performed in positive and negative ESI mode, respectively, using the multiple reaction monitoring (MRM) mode with three transitions for each analyte to enhance the specificity of the method and avoid false positives. The methodology reported in this work is capable of detecting residues of glyphosate and AMPA in soybean samples with limits of quantification of 0.30 and 0.34 mg kg^{-1}, respectively. This alternative method has throughput advantages such as simpler sample preparation and faster chromatographic analysis. Copyright © 2009 John Wiley & Sons, Ltd.

Glyphosate (*N*-phosphonomethyl glycine) is a widely used herbicide, which rapidly degrades into its aminomethylphosphonic acid (AMPA) metabolite.[1] It has a broad spectrum of activity, and it is very effective, even on plant roots, with little harmful effects on mammals. Its high efficacy, low toxicity and low cost, compared with those of other herbicides, have led to its wide utilization in several crops. Due to its low toxicity, the maximum residues levels (MRLs) for glyphosate established around the world are generally greater than those set for other pesticides. For example, the Codex Alimentarius Commission[2] established the MRL as 0.2 mg kg^{-1} for glyphosate in soybean, and the National Health Surveillance Agency (ANVISA) in Brazil[3] set the MRL as 10 mg kg^{-1}.

The physical and chemical properties of glyphosate and AMPA, such as their involatility, high water solubility and absence of chromophores, make them very difficult to analyze without derivatization using gas or liquid chromatography with traditional detectors. They may, however, be analyzed by mass spectrometry (MS) or tandem mass spectrometry (MS/MS), especially using electrospray ionization (ESI), because they are very polar and thus easily ionized by this technique. Only a few methods using MS or MS/MS coupled with liquid chromatography have been reported for glyphosate analysis. In most reported methods, pre- and post-column derivatization procedures were employed to produce derivatives suitable for fluorescence detection.[4,5] Vreeken and co-workers analyzed glyphosate, AMPA, and glufosinate in water using reversed-phase liquid chromatography (LC) separation after pre-column derivatization with 9-fluorenylmethoxycarbonyl chloride and detection by MS/MS.[6] Bauer *et al.*[7] determined glyphosate and AMPA in water samples using ion chromatography followed by ESI-MS detection with a single quadrupole mass spectrometer. Ion chromatography separation without derivatization was also used by Granby and co-workers after clean-up on a reversed-phase column to analyze glyphosate and AMPA through ESI-MS/MS.[8,9] Goodwin and co-workers[10,11] studied the fragmentation pathway of glyphosate and AMPA using an ion trap mass spectrometer. Other reported methods for the analysis of glyphosate include capillary electrophoresis,[12–14] ion chromatography with conductivity,[15] ion chromatography with fluorescence,[16–18] gas chromatography,[19–21] immunoassays,[22,23] nuclear magnetic resonance,[24] and integrated pulse amperometry.[25]

In this work, we investigated the potential of reversed-phase LC/MS/MS for the quantification of glyphosate and AMPA in spiked soybean samples. The compounds were analyzed without derivatization using calibration curves prepared in the matrix after a simple protein precipitation

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step to minimize the complexity of the sample. The mobile phase composition and the matrix effects were also investigated to validate the method using a high-flow gradient program in a total run time of 4 min for each analyte.

EXPERIMENTAL

Chemicals and standards

Glyphosate and AMPA standards were obtained from Sigma-Aldrich (Steinheim, Germany). Methanol and dichloromethane (HPLC-grade) were purchased from J. T. Baker (Deventer, The Netherlands) and ammonium carbonate P.A. was obtained from Merck (Darmstadt, Germany). Purified water was obtained on an EASYpure RF system from Barnstead (Dubuque, IA, USA).

The stock solutions of glyphosate and AMPA at 500 mg L^{-1} were prepared by dissolution of the standards in water. The solutions were maintained at 4°C protected from light and kept in polypropylene tubes to avoid adsorption to glass. The calibration standards were diluted in water or blank soybean extracts for the calibration curves.

High-performance liquid chromatography (HPLC)

An Agilent 1100 series (Agilent Technologies, Waldbronn, Germany) liquid chromatography system was operated at flow rate of 1.2 mL min^{-1} without split using a Zorbax Eclipse® RDB C8 (Agilent Technologies) analytical column (150 mm × 4.6 mm (i.d.), 5 µm particle size), maintained at 25°C during the experiments. The volume injected into the LC/MS/MS system was 40 µL. The binary mobile phase consisted of water with 1.5 mmol L^{-1} of ammonium carbonate (phase A) and methanol/water (95/5%) with 1.5 mmol L^{-1} of ammonium carbonate (phase B). The initial composition of 65% A and 35% B (v/v) was held for 0.5 min, followed by linear ramping to 90% of B over 1.75 min. After ramping, the mobile phase was returned to the initial composition, and this was held for 1.75 min. The total chromatographic run time was 4.0 min.

Mass spectrometry

The experiments were performed using an API 4000™ (Applied Biosystems/MDS Sciex, Concord, Canada) triple quadrupole mass spectrometer with a TurboIonSpray® (electrospray) ionization source. The capillary voltage was maintained at 5500 and −4500 V for positive and negative ion modes, respectively, and the temperature of the turbo heaters was set at 750°C. Air was used as the nebulizer gas (GS1) at a pressure of 55 psi and the heater gas (GS2) was also at 55 psi. Nitrogen was used as the curtain gas at a pressure of 12 psi in the interface and as the collision gas (CAD Gas) at 10 arbitrary units in the collision cell (Q2). The declustering potential (DP), collision energy (CE), and collision cell exit potential (CXP) parameters used in the multiple reaction monitoring (MRM) mode are presented in Table 1. The dwell time was set at 150 ms and the pause time at 5 ms. The data were acquired using Analyst® software (Applied Biosystems).

Sample preparation

Organic soybeans were purchased from a local market in São Paulo City and used as blank samples. The soybean samples were blended and extracted immediately or stored at −4°C. The blended sample (2.0 g) was weighed in a 50 mL polypropylene tube and extraction was carried out with 20.0 mL of water and 5.0 mL of dichloromethane by mechanical agitation with a shaker for 60 min. After the extraction, the sample was centrifuged for 15 min at 3000 rpm and an aliquot of 1.0 mL was transferred into a 15 mL polypropylene tube followed by addition of 1.0 mL of methanol for protein precipitation. The contents of the tube were mixed by vortex for 1 min and centrifuged at 3000 rpm for 15 min. Prior to analysis, the samples were diluted 50-times by mixing 20 µL of the extract into a 2 mL chromatographic vial with 980 µL of water. Then, 40 µL of sample was injected into the LC/MS/MS system.

Table 1. Optimized MRM detection parameters

Analyte	Transition (m/z)	DP (V)	CE (eV)	CXP (V)
	110 > 63	−41	−26	−1
AMPA	110 > 79	−41	−38	−3
	110 > 81	−41	−18	−3
	170 > 42	50	37	8
Glyphosate	170 > 60	50	25	10
	170 > 88	50	15	14

Validation of the method

The method was validated according to 2002/657/EC.[26] Four batches of spiked and blank samples were analyzed to evaluate the performance of the method. These batches were analyzed on four consecutive days using a total of 83 soybean samples. The spiked levels were 0.2 and 0.6 mg kg^{-1} ($n = 18$), 0.4 mg kg^{-1} ($n = 28$) and 0.8 and 2.0 mg kg^{-1} ($n = 3$). Thirteen blank samples were also analyzed. The intraday results for glyphosate and AMPA are shown in Tables 2 and 3, respectively. In Table 4, the results from the four validation days were pooled to evaluate the interday recovery and precision.

Table 2. Results of glyphosate in spiked soybean samples

Batch/Day	Spike level (mg kg^{-1})	Recovery (%)	SD (%)	RSD (%)	n
	0.2	88.1	11.5	13.0	6
	0.4	93.0	8.3	9.0	6
1	0.6	94.9	5.7	6.0	6
	0.8	102.4	—	—	1
	2.0	91.6	—	—	1
	0.2	76.9	9.9	12.9	6
	0.4	89.2	8.1	9.0	6
2	0.6	85.8	6.1	7.1	6
	0.8	89.7	—	—	1
	2.0	89.9	—	—	1
	0.2	73.9	7.9	10.8	6
	0.4	80.9	7.4	9.2	6
3	0.6	85.1	4.5	5.3	6
	0.8	84.2	—	—	1
	2.0	85.7	—	—	1
4	0.4	109.1	6.6	6.1	10

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Table 3. Results of AMPA in spiked soybean samples

Batch/Day	Spike level (mg kg^{-1})	Recovery (%)	SD (%)	RSD (%)	*n*
	0.2	93.1	7.7	8.3	6
	0.4	95.5	5.5	5.8	6
1	0.6	87.9	7.5	8.5	6
	0.8	78.3	—	—	1
	2.0	91.1	—	—	1
	0.2	87.7	12.8	14.5	6
	0.4	85.8	8.0	9.3	6
2	0.6	95.6	8.3	8.7	6
	0.8	85.7	—	—	1
	2.0	91.8	—	—	1
	0.2	88.2	4.9	5.5	6
	0.4	73.9	4.2	5.7	6
3	0.6	81.3	6.1	7.5	6
	0.8	93.1	—	—	1
	2.0	89.9	—	—	1
4	0.4	94.0	5.6	6.0	10

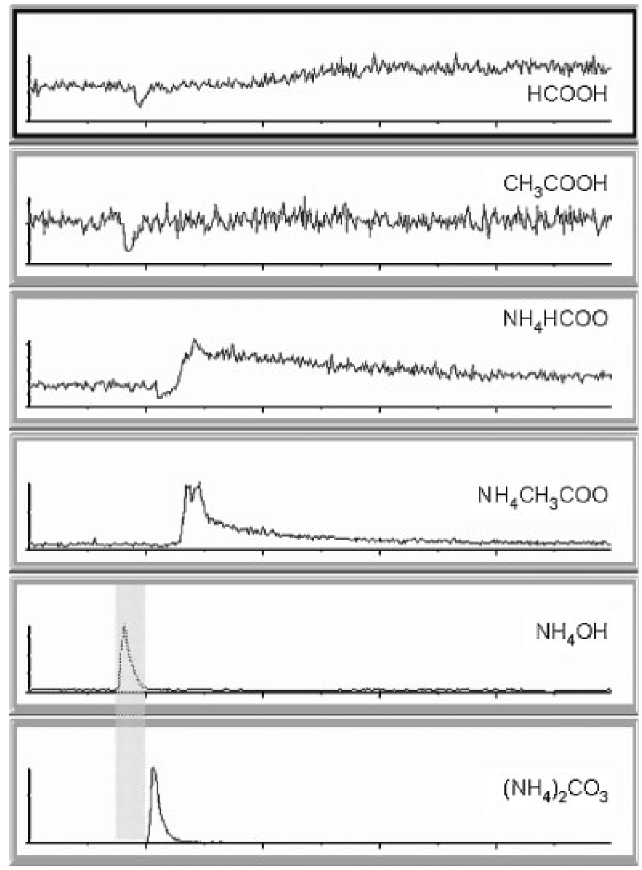

Figure 1. MRM signals for glyphosate (*m/z* 170 > 88) retained on a C8 reversed-phase column using different buffers in the mobile phase.

RESULTS AND DISCUSSION

Optimization of the method

Glyphosate and AMPA can be ionized in both positive and negative ESI modes, by addition or loss of a proton, respectively. However, in the MRM experiments, higher sensitivity was achieved for glyphosate in the positive ion mode and for AMPA in the negative ion mode. For this reason, the individual samples were injected twice in each analytical run. The MRM methods were optimized using three precursor ion > product ion transitions for each analyte, where one transition was employed for quantification purposes and the other two transitions for confirmation. For quantification, the *m/z* 170 > 88 and 110 > 79 transitions were used for glyphosate and AMPA, respectively. The confirmation transitions for glyphosate were *m/z* 170 > 60 and 170 > 42, and those for AMPA were *m/z* 110 > 63 and 110 > 81.

Due to the strong interactions of glyphosate and AMPA with the free silanol groups from the stationary phase, the better buffers used in the chromatographic elution were those operating at high pH. Figure 1 shows the different behavior of glyphosate using traditional buffers for LC/MS/MS. Although ammonium carbonate and ammonium hydroxide showed better peak shapes than other buffers, ammonium carbonate was chosen as it gave longer retention times than ammonium hydroxide. The shaded portion in Fig. 1 highlights the difference in the retention time of glyphosate for both buffers. The pH values of the aqueous mobile phase (phase A) using ammonium carbonate and ammonium hydroxide were 9 and 10, respectively.

The sensitivity of the analysis was better when a high chromatographic flow rate was used. Even when starting the run with 65% of aqueous phase, no loss of ionization efficiency was observed, and this solvent composition provided the best analyte peak shapes. A compromise was thus reached between sensitivity and analysis time, producing column pressure of less than 200 bar. Because of the low retention times of the analytes on the reversed phase, some samples were analyzed using direct flow injection analysis (FIA) into the tandem mass spectrometer without the C8 analytical column. This approach gave worse peak shapes and lower accuracy because of matrix suppression. Therefore, the use of a reversed-phase column was important to remove some non-polar background compounds from the mobile phase and to enhance the performance of the method.

Table 4. Pooled interday results for glyphosate and AMPA in spiked soybean samples

Compound	Spike level (mg kg^{-1})	Recovery (%)	SD (%)	RSD (%)	*n*
	0.2	79.6	9.8	12.2	18
	0.4	109.1	6.6	6.1	28
Glyphosate	0.6	88.6	5.4	6.1	18
	0.8	92.1	—	—	3
	2.0	89.0	—	—	3
	0.2	89.7	8.5	9.4	18
	0.4	94.0	5.6	6.0	28
AMPA	0.6	88.3	7.3	8.2	18
	0.8	85.7	—	—	3
	2.0	90.9	—	—	3

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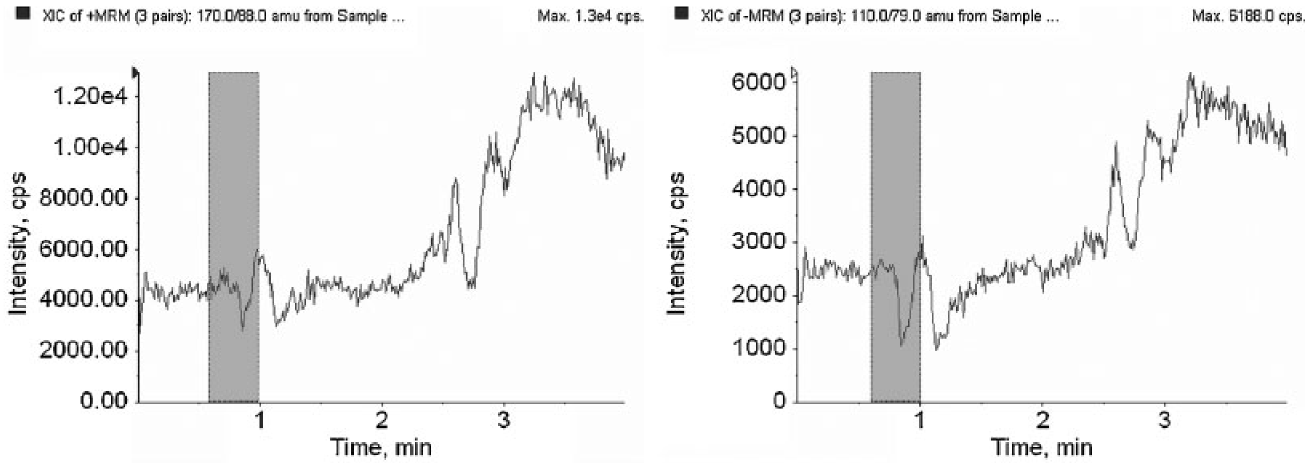

Figure 2. MRM signals obtained for post-column infusion of glyphosate (left) and AMPA (right). The shaded portion shows the signal suppression for the compounds at their retention times.

Matrix effect

Three different solvents were applied for the protein precipitation of the dirty extracts obtained from the soybean aqueous extract: acetone, acetonitrile and methanol. The use of acetone and acetonitrile as solvents reduced the recovery of the analytes because of co-precipitation with the proteins in the samples. The best recoveries for glyphosate and AMPA were obtained using methanol as a solvent; however, the use of protein precipitation did not eliminate the matrix interference, and some signal suppression was observed for both analytes. Solid-phase extraction and liquid-liquid partition were also used to purify the samples, but ion suppression could not be completely avoided. Figure 2 shows the chromatograms obtained using the post-column infusion method for the evaluation of the matrix effects.

Although glyphosate and AMPA detection showed noticeable signal suppression effects even after liquid-liquid partition and protein precipitation, as shown in Fig. 2, the large dilution volume applied to the samples was effective in minimizing this effect in the ESI source. In order to overcome interference caused by the matrix effect, the analytes were quantified using matrix-matched calibration standards. The standards of the compounds were prepared by dilution of the analytes in soybean blank sample extracts at concentrations of 0.1, 0.2, 0.5, 1.0, 1.5 and 3.0 $\mu g\,L^{-1}$.

Sensitivity

Although glyphosate and AMPA, with their low molecular weight and high polarity, are less sensitive in ESI-MS/MS than other pesticide compounds, low limits of detection were obtained. The limit of detection (LOD) and the limit of quantification (LOQ) were defined as signal-to-noise (S/N) ratios of 3:1 and 10:1, respectively. For the quantification transitions, the LOD and LOQ were 0.09 and 0.30 $mg\,kg^{-1}$ for glyphosate, and 0.1 and 0.34 $mg\,kg^{-1}$ for AMPA, respectively; they were calculated using a matrix-matched standard solution at a concentration of 0.2 $\mu g\,L^{-1}$ (Fig. 3).

Because of the low retention time of the compounds in reversed-phase columns, three MS/MS transitions were monitored to enhance the reliability of the data and to avoid false positives. Figures 4 and 5 show the S/N ratios obtained for the analytes at three transitions in a sample spiked at 0.4 $mg\,kg^{-1}$.

The calibration was performed using external matrix-matched calibration solutions ranging from 0.1 to 3.0 $\mu g\,L^{-1}$. For glyphosate (m/z 170 > 88) and AMPA (m/z 110 > 79), the correlation coefficients (r^2) calculated by weighted regression ($1/x$) were 0.9991 and 0.9998 and the slopes of the calibrations performed on the matrix-matched standards were less than those obtained on the solvent standards.

Recovery and performance

As glyphosate and AMPA have high solubility in water (up to 12 $mg\,L^{-1}$), aqueous extraction was successfully applied to recover these analytes from the soybean spiked samples. Liquid-liquid partitioning with dichloromethane and protein precipitation were used to purify the extract, yielding the recovery results presented in Tables 2, 3 and 4.

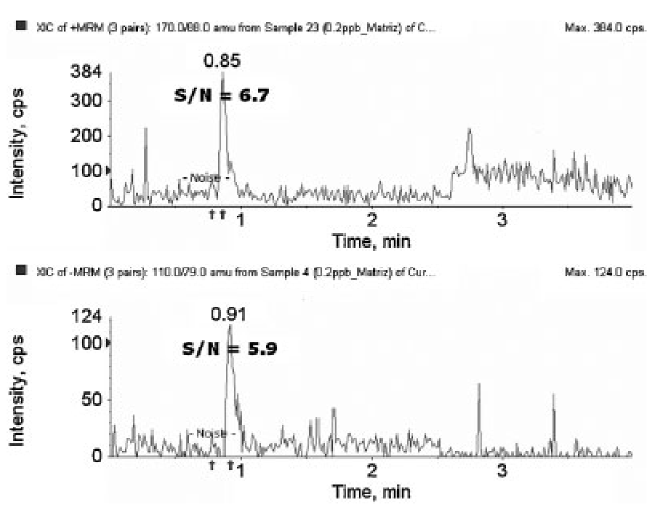

Figure 3. MRM signals obtained from standard samples at concentrations of 0.2 $\mu g\,L^{-1}$ (0.2 $mg\,kg^{-1}$) for glyphosate (m/z 170 > 88) (top) and AMPA (m/z 110 > 79) (bottom).

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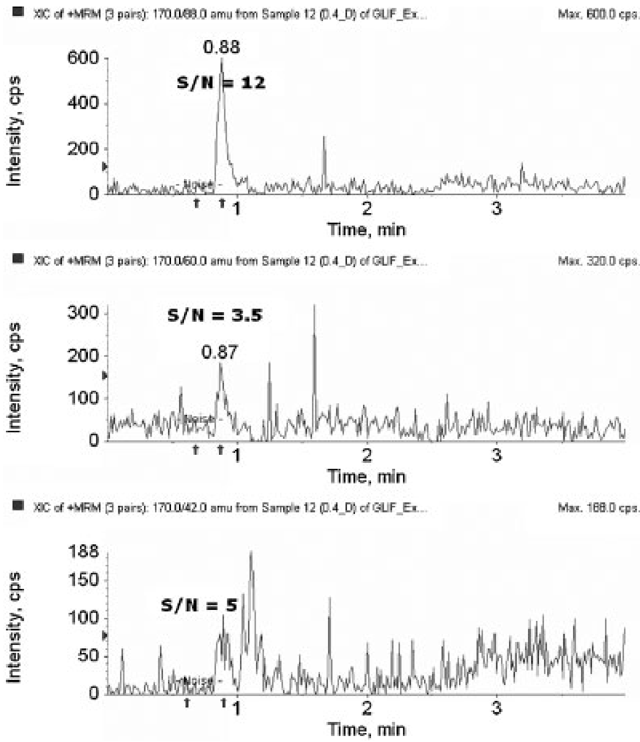

Figure 4. MRM signals of sample spiked with glyphosate at a concentration of $0.4\,mg\,kg^{-1}$.

The pooled results presented in the Table 4 show that the recoveries for spiked concentrations of 0.2, 0.4 and $0.6\,mg\,kg^{-1}$ were between 79.6 and 109.1% with relative standard deviation (RSD) of less than 12.2% for both analytes. According to the European Community Directive 657/2002/EC,[26] for concentrations higher than $0.01\,mg\,kg^{-1}$,

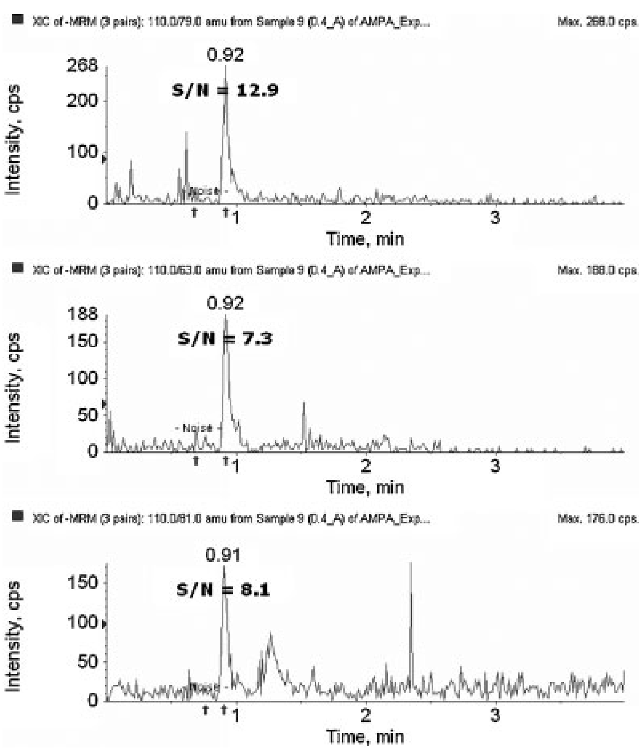

Figure 5. MRM signals of sample spiked with AMPA at a concentration of $0.4\,mg\,kg^{-1}$.

the trueness (measured through recovery) of quantitative methods should be between 80 and 110%.

The decision limit (CCα) and detection capability (CCβ) established by the European Community through the Directive 657/2002/EC were, respectively, 0.03 and $0.05\,mg\,kg^{-1}$ for glyphosate and 0.03 and $0.06\,mg\,kg^{-1}$ for AMPA. The curves for the control samples showed correlation coefficients higher than 0.9885 for both analytes at the studied spike levels.

CONCLUSIONS

An alternative methodology for the determination of glyphosate and AMPA residues in soybean with simpler sample preparation, fast chromatographic analysis and sensitive detection was presented. The sensitivity and specificity of the method are suitable to meet the residue limits established in most countries for evaluating the presence of glyphosate and its major metabolite in soybean samples. The method was developed and validated using spiked organic soybean samples according to the Directive EC/2002/657. The presented methodology improves and simplifies routine glyphosate quantitative analysis throughput, by combining the power of liquid chromatography with the sensitivity and specificity provided by tandem mass spectrometry. The presented results suggest that this technique may also be used to analyze residues of these compounds in other crop matrices, such as corn and cotton, where glyphosate is widely used.

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